

Exercise Sheet Generative Models

Exercise 1: Global Optimality of the GAN objective (10 + 10 + 20 P)

In this exercise, we want to show that the global optimal solution for the minimax game

$$\min_G \max_D V(D, G) = \min_G \max_D \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}(\mathbf{x})} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_{\mathbf{x}}(\mathbf{z})} [\log(1 - D(G(\mathbf{z})))] \quad (1)$$

for training Generative Adversarial Networks is that the data distribution gained from sampling from p_g is equal to the real data distribution p_{data} .

(a) Therefore, we first consider the optimal discriminator D for any given generator G . Show that for fixed G , the optimal discriminator D is

$$D_G^*(\mathbf{x}) = \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})} \quad (2)$$

Hint: For any $(a, b) \in \mathbb{R}^2 \setminus \{0, 0\}$, $y \in [0, 1]$, the function $f(y, a, b) = a \log(y) + b \log(1 - y)$ achieves its maximum at $\frac{a}{a+b}$.

(b) Show that the maximum $C(G) = \max_D V(G, D)$ of the training criterion can be reformulated to:

$$C(G) = \mathbb{E}_{\mathbf{x} \sim p_{\text{data}}} \left[\log \frac{p_{\text{data}}(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})} \right] + \mathbb{E}_{\mathbf{x} \sim p_g} \left[\log \frac{p_g(\mathbf{x})}{p_{\text{data}}(\mathbf{x}) + p_g(\mathbf{x})} \right] \quad (3)$$

(c) Show that the global minimum of $C(G)$ is $C^* = -\log(4)$ and that reaching it is equivalent to $p_g = p_{\text{data}}$. Hint: Use the fact, that the the Jensen Shannon Divergence $JSD(P\|Q) = \frac{1}{2} (KL(P\|M) + KL(Q\|M))$ is always positive.

Exercise 2: Reformulating the loss function of diffusion models (20 P)

(a) Show that

$$L_{vlb} = \mathbb{E}_q \left[-\log \frac{p_{\theta}(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \right]$$

can be reformulated to:

$$L_{vlb} = L_0 + L_1 + \dots + L_{T-1} + L_T$$

where

$$\begin{aligned} L_0 &= -\log p_{\theta}(x_0 | x_1) \\ L_{t-1} &= D_{KL}(q(x_{t-1} | x_t, x_0) \| p_{\theta}(x_{t-1} | x_t)) \\ L_T &= D_{KL}(q(x_T | x_0) \| p(x_T)) \end{aligned}$$

with the help of the Markov assumption in Diffusion models.

Exercise 3: Programming (40 P)

Download the programming files on ISIS and follow the instructions.
 Generate MNIST with diffusion model in PyTorch.